Ollscoil Mhá Nuad
Maynooth University

An tAcadamh Taighde Iarchéime
Graduate Research Academy

Maynooth University
National University of Ireland Maynooth

Form No. GSF1
(Version 6, September 2026)

Initial Meeting Record (IMR): Research & Professional Development Plan

Principles: A key purpose of the initial meeting between a research student and academic supervisor(s) is to ensure that expectations around supervision have been agreed and that an initial plan for the research and professional development programme has been set out. During this meeting, the student and supervisor will identify modules that will enhance the student's research programme, agree on a calendar of meetings, and ensure that the student is aware of and agrees to be bound by the University regulations. Additionally, this meeting allows the supervisor and student to document the initial plan and scope of the research project.

Details of the agreed programme and professional development plan are recorded below and are then reviewed and approved by the Departmental Research Student Progress Committee (DRSPC), which retains a copy in the department. The IMR informs the annual progress review subsequently conducted by the student's DRSPC.

Student Details

Last Name:	Kiely	First Name:	Jerry
Student Number:	26266311	Full-time/Part-time	Full-time
Department:	Computer Science	MU Email:	jeremiah.kiely.2027@mumail.ie

Supervisory and Programme Details

Supervisor:		Abeer Eshra	
Co-Supervisor *(if applicable)*:			
Supervisory Team *(if applicable)*:			
External Supervisor *(if applicable)*:			
Will you have a remote study arrangement for some or all of your research programme? If so, please state location and duration		No	
If Joint PhD, please state name of partner institution:			
Year admitted:	2026	Term admitted: (semester 1/semester2)	Semester 1
Expected date of submission:			
Indicate the thesis submission format you plan to pursue:	☑ **PhD by Thesis**	☐ **PhD by Publication**	☐ **Practice-Engaged / Performance-Based**

Initial Meeting Record (student to complete)

Did you attend the University/GRA Induction Programme?	Yes	X	No	
Have you received a formal letter of offer from the Admissions Office and agree the contents are correct?	Yes	X	No	
Do you agree to be bound by the Maynooth University regulations?	Yes	X	No	
Did you receive a copy of the Departmental Graduate Handbook?	Yes	X	No	
If applicable, did you attend the Departmental Induction Programme?	Yes	X	No	
If Joint PhD, have you read and understood the Memorandum of Agreement governing your Joint PhD programme?	Yes	N/A	No	N/A
Have you completed the *Expectations of Research Supervision Worksheet* and discussed with your supervisor? (See Appendix)	Yes	X	No	

Research Plan

Provisional Title of Thesis

Provisional Title of Thesis / Area of Research:

Provisional title: Design, Modelling and Experimental Validation of Programmable Biomolecular Computing Systems
Area of research: Computer Science - DNA Computing

Goals and Objectives

List specific research goals and/ or research questions: 500 words max.

The goal of my research is to develop models and simulations of the scaffolded DNA computing (SDC) process, to validate these models through experimentation, and to gain valuable insights of the SDC process through these models and simulations, which will inform the development of libraries of motifs to support renewable DNA computing.

O1 - Theory of the Scaffolded DNA Computing process:

1. Modeling the SDC process - kinetic, thermodynamic
2. Developing simulations (stochastic, monte carlo, etc.) and numerical computational approaches
3. Investigating factors that influence scalability of SDC - thermodynamic, stoichiometric, etc.
4. Validation of models against existing models, theory, and state of the art
5. Validation of models against experiments conducted in the wet lab

O2 - Design of Renewable SDC Library:

1. Design an ensemble of thermodynamically regulatable motifs (tiles)
2. Characterize said motifs computationally
3. Incorporate said motifs into a reusable library to support renewable DNA computing

Professional Development Plan

Career path

Outline potential career paths:

After completing the PhD to continue in academia as a post-doc researcher, and to give back to the research community. During the post-doc research period to investigate available avenues of funding, and to create my own research group. Also to offer consultancy to industry and to collaborate with other research groups.

Skills: Please review the IUA PhD Graduate Skills Statement

Main skills to develop:

Research Skills:

- become familiar with the state of the art
- conduct further research in related fields (such as statistical mechanics, number theory)
- develop the ability to review technical papers
- enhance my time management skills
- become familiar with the various funding sources, and grant application processes

Personal Effectiveness Transferable Skills:

- learn financial management (budgeting) skills
- enhance presentation skills

Team-working and leadership skills

- gain further experience working in teams

Communication skills

- gain experience writing research and technical papers
- gain experience communicating ideas effectively

Outline of Modules and Courses

Provisional plan of modules/courses to be completed during the programme; Note that PhD students must complete a minimum of 30 credits over the duration of the programme, 15 subject-specific modules and 15 transferable skills modules. Research Master's students must complete a minimum of 10 credits over the duration of the programme, 5 subject-specific and 5 transferable modules.

Module Code	Module Name	Subject-specific or transferable?	Number of credits	Year of study in which module to be taken
CS805	The Mathematics & Theory of Computer Science	CS	5	1
CS900	Computer Science 1	CS		1
CS865	Computer Science Phd Year 1	CS	15	1

Other Courses/ Workshops:

MP461	Statistical Mechanics	

Proposed Frequency of Meetings for the Academic Year with Principal Supervisor(s)

Frequency	**Venue**
Structured weekly meeting with flexibility to be reached on a daily basis.	HI (The Hamilton Institute)

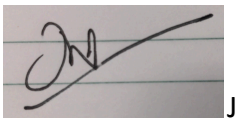
Please discuss the requirements for attendance on campus with your supervisor and specify below if you will be based on campus daily, working in a laboratory/archive or if undertaking the research in an off-site institution or elsewhere. If in receipt of research funding, please ensure compliance with any terms and conditions as regards attendance and/or residency:

Details of location for research:

The student will be based at the Hamilton Institute and has been allocated dedicated research desk space there. The majority of the research will be computational and theoretical and will be carried out using the student's funded laptop and the research facilities available at the Hamilton Institute, and access to GPU computing resources within the Department of Computer Science.

The student will also have access to the wet-lab facilities, where required, for experimental validation of computational models and results. Attendance on campus will therefore be primarily based at the Hamilton Institute, with laboratory access used as needed throughout the project.

Agreement Signatures

Student:	 Jerry Kiely	Date:30th October 2026
Supervisor:	Abeer Eshra	Date: 30/10/2026
Supervisor:	N/A	Date
If research programme involves more than one department or a Joint PhD, please ensure that the supervisors from both departments confirm agreement.		
Joint Supervisor:	N/A	Date

Approved by members of Departmental Research Student Progress Committee

Name and Signature:	Date:
Name and Signature:	Date:
Name and Signature:	Date:
Name and Signature:	Date:

Once approved by the DRSPC, the form should be sent by the Chair of the DRSPC to the student and supervisor(s) and a copy kept in the department.

Appendix

Expectations of Research Supervision

Supervisors and research students often have differing expectations of supervision. The supervisor and student should complete this scale independently and then compare their responses. Circle a number depending on whether you think the responsibility lies more with the supervisor or the student.

	Supervisor	Rating	Student
1.	It is the supervisor's responsibility to select the research topic.	1 2 3 4 5	The student is responsible for selecting her/ his own topic.
2.	The supervisor should decide which theoretical framework and/or methodology is most appropriate.	1 2 3 4 5	The students should decide which theoretical framework and/or methodology they wish to use.
3.	The supervisor should develop an appropriate program and timetable of research and study for the student.	1 2 3 4 5	The student should develop their own program and timetable of research.
4.	The supervisor should ensure that the student has access to the resources and facilities that they need.	1 2 3 4 5	It is the student's responsibility to locate and access all necessary resources and facilities.
5.	It is the supervisors' responsibility to advise the student of all relevant policies, procedures and requirements.	1 2 3 4 5	The student is responsible for being familiar with the relevant policies, procedures and requirements.
6.	It's up to the supervisor to build an appropriate professional relationship with the student.	1 2 3 4 5	It is the student's responsibility to manage the relationship with their supervisor.
7.	The supervisor should insist on regular meetings with the student.	1 2 3 4 5	The student should decide when she/he wants to meet.
8.	The supervisor should check regularly that the student is working consistently and on track.	1 2 3 4 5	The student should work independently and not have to account for how and where time is spent.
9.	It is the supervisor's responsibility to should ensure that the thesis is finished by the required submission date.	1 2 3 4 5	It is up to the student to finish within the allocated timeframe.
10.	The supervisor should insist on seeing all drafts of work to ensure that the student is on the right track.	1 2 3 4 5	It's up to the student to decide when to show drafts of their work.
11.	The supervisor should assist in the writing, editing and presentation of the thesis.	1 2 3 4 5	The writing of the thesis should only be the student's own work and they must take full responsibility for presentation of the thesis.
12.	The supervisor is responsible for deciding when the thesis can be submitted for examination.	1 2 3 4 5	The student should decide when they are ready to submit the thesis for examination.

Adapted from: Moses, I. (1985). Supervising postgraduates. HERDSA Green Guide No 3, Kensington: Higher Education Research & Development Society of Australasia.

www.ithinkwell.com.au **Supervising PhD Students, Kearns and Finn, 2017.**